SIDDHARTH DIALANI

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Digital Twin & Simulation Engineer | AI-Driven 3D Workflows

- Digital-twin and simulation practitioner with an MBA focus in AI and Business Analytics and hands-on project experience in 3D scene preparation, BIM/USD interoperability, environmental visualization, and spatial reconstruction. Experienced in exploring workflows across Blender, Revit, Unreal Engine, NVIDIA Omniverse, Isaac Sim, and OpenUSD. Skilled in investigating cross-platform asset integration, AI-assisted automation, and simulation workflows to support interactive 3D environments. Interested in developing practical digital-twin solutions that connect 3D modelling, software integration, and simulation for industrial, urban, and built-environment applications.

SKILLS

- **Digital Twin & 3D:** Blender 5.2, Unreal Engine, NVIDIA Omniverse, Isaac Sim, Revit, OpenUSD/USD Bridge
- **Reconstruction & Visualization:** COLMAP, Gaussian Splatting, SuperSplat, iPhone LiDAR/RoomPlan, 360° capture, PlayCanvas
- **Integration & Simulation:** MCP, Revit MCP, USD exchange workflows, OpenXR, Meta Quest 3 PC Link, locomotion, collision/physics testing
- **Programming & Tools:** Python, JavaScript, TypeScript, React, Next.js, FastAPI, Node.js, Codex, Claude Code, Cursor, n8n, GitHub, Docker, Vercel, PostgreSQL
- **Languages:** English (Fluent), Hindi (Native), Marathi (Conversational)

WORK EXPERIENCE

MIDDLESEX UNIVERSITY DUBAI Sep 2026 – Present

Student Research Assistant (SRA) Dubai, UAE

- Research generative-AI and computer-vision workflows for digital twins, 3D scene reconstruction, and spatial computing.
- Build and test Blender, iPhone LiDAR/RoomPlan, 360° capture, COLMAP, Gaussian Splatting, SuperSplat, PlayCanvas, and OpenUSD workflows.
- Explore scene-preparation automation and MCP integrations across Revit, Omniverse, and Isaac Sim; document interoperability and validation issues.

CERTIFICATIONS

ML Practitioner Certificate — Dataiku (July 2026)
Core Designer Certificate — Dataiku (July 2026)

PROJECTS

Dubai Urban & Environmental Digital Twin 2026

- Built and deployed a browser-based building-environment simulation for a Dubai site, with time-of-day scenarios, surface-temperature heat maps, and panels for temperature, humidity, solar radiation, and wind. Uses deterministic simulated inputs, not live weather or sensor data.

Technologies: Environmental simulation, heat-map visualization, scenario controls, Vercel

HeatShield — Accommodation Twin 2026

- Built and deployed HeatShield as a browser-based accommodation-twin prototype, establishing a dedicated web interface for the accommodation-focused digital-twin experience.

Technologies: Web application, accommodation-twin interface, Vercel

AI-Powered SimReady Robotics Workflow 2026 – Ongoing

- Developing an AI-assisted workflow to discover Hugging Face SimReady assets and prepare them for robotics and digital-twin scenes; exploring OpenUSD scene assembly and simulation preparation in NVIDIA Isaac Sim.

Technologies: Hugging Face SimReady, NVIDIA Isaac Sim, OpenUSD/USD, Python, Codex

Blender → USD → Unreal / Omniverse 2026

- Explored cross-application workflows for transferring 3D scenes and assets from Blender through OpenUSD to Unreal Engine and Omniverse. Used Blender Python automation and USD conversion to investigate scene structure and compatibility.

Technologies: Blender, Python, OpenUSD/USD, Unreal Engine, NVIDIA Omniverse

EDUCATION

MBA, Business Analytics & Artificial Intelligence • Middlesex University Dubai Expected Feb 2027
BCA • BYK College of Commerce, Nashik, India